CHAPTER 3

The Multi-Curve World: OIS, SOFR, and the Death of LIBOR

3.1 Why there used to be one curve, and why there no longer is

Before 2007, the rates market was, for pricing purposes, simple: LIBOR was both the rate banks borrowed at and the rate used to discount future cash flows, so a single curve did both jobs. The 2008 financial crisis broke this. Banks stopped trusting each other's unsecured credit, LIBOR (which embeds interbank credit and liquidity risk) diverged from the rate on fully collateralized transactions, and the market permanently split into two roles that a single curve can no longer play at once.

Key result: discounting curve vs. forwarding curve

A **discounting curve** answers "what is a certain future cash flow worth today?" and must reflect the rate at which the specific cash flow is actually funded — for a collateralized derivative, the rate paid on posted collateral, which is an overnight risk-free rate. A **forwarding curve** (or **projection curve**) answers "what rate does this floating leg reset to?" and must match the index the contract actually references. Since the 2008 crisis, these are built as separate curves and are, in general, numerically different.

For a collateralized USD swap today, discounting is done off the **SOFR OIS curve** — the curve implied by overnight-indexed swaps referencing the Secured Overnight Financing Rate — while a SOFR-referencing floating leg is projected off a SOFR forward curve. In the current all-SOFR world these two curves largely coincide for USD, which is a genuine simplification relative to the old LIBOR/OIS multi-curve setup; but for legacy products, cross-currency structures, and any market still running its own credit-sensitive benchmark, the discounting and forwarding curves remain distinct, and a pricing system must be built to keep them so.

PITFALL

Even in an ostensibly "single curve" SOFR world, treating discounting and forwarding as automatically identical is a trap. SOFR OIS and SOFR term structures built from futures versus swaps can imply subtly different curves depending on the instrument set and interpolation method used to build each; and the moment a cross-currency swap, a legacy LIBOR fallback, or a credit-sensitive alternative rate enters the picture, the two curves diverge again, sometimes by a economically significant amount. A calibration engine that hard-codes "one curve does both jobs" will silently mis-discount the moment it is pointed at a book that isn't purely vanilla single-currency SOFR.

3.2 LIBOR's cessation and what replaced it

USD LIBOR panels stopped publishing representative rates after June 30, 2023, following a multi-year transition process coordinated by the Alternative Reference Rates Committee (ARRC) in the US and equivalent bodies elsewhere. The replacement benchmarks are **risk-free**

rates (RFRs): overnight rates, backed by large volumes of real transactions, administered independently of any bank panel submission.

Key result: the major RFRs

- **USD — SOFR** (Secured Overnight Financing Rate): overnight Treasury repo rate, published by the New York Fed, backed by well over $1 trillion of daily transaction volume.
- **GBP — SONIA** (Sterling Overnight Index Average): overnight unsecured rate, published by the Bank of England.
- **EUR — €STR** (Euro Short-Term Rate): overnight unsecured rate, published by the ECB.

All three share a structural feature that LIBOR did not: they are *overnight* rates. There is no natively-published 3-month or 6-month SOFR the way there was a natively-submitted 3-month LIBOR. Any longer-tenor "SOFR rate" is a construction built from the overnight rate, and *how* it is constructed is the single most consequential convention question in the post-LIBOR market.

3.3 Forward-looking versus backward-looking term rates

There are two structurally different ways to build a usable 1-, 3- or 6-month rate out of a published overnight series, and the difference between them is not cosmetic — it changes what a caplet on the rate even *is*.

Compounded-in-arrears (backward-looking) rates compound the actual overnight fixings realised *during* the accrual period:

$$1 + \tau \cdot L^{\text{arrears}}(T_1, T_2) \;=\; \prod_i (1 + \tau_i\, r_{o,i})$$

where the product runs over each business day i in $[T_1, T_2]$ and $r_{o,i}$ is that day's overnight fixing. The rate is only known in full once the period has *ended* — it has genuine path-dependence within the accrual period, closer in character to an Asian (average-rate) option than to the simple LIBOR-style payoff it replaced. This construction — Daily Simple SOFR or Compounded SOFR in arrears — is by far the dominant convention in the SOFR swaps, caps and floors market as of 2026.

Term SOFR (forward-looking) rates, by contrast, are published in advance, derived from SOFR futures prices, and behave exactly like old LIBOR: known at the start of the period, applied at the end. CME Term SOFR is the main published version. Its use, however, is deliberately restricted by convention and by ARRC-recommended best practice, primarily to end-user-facing cash products such as business loans, precisely to avoid re-concentrating the market's risk in a rate that (like LIBOR) is derived rather than directly transacted.

Key result: the payoff difference that matters for pricing

A cap on a *forward-looking* term rate settles a payoff known entirely at the start of its accrual period — structurally identical to a LIBOR caplet, and the classical Black-formula machinery (Chapter 6) applies with essentially no modification beyond swapping the index.
A cap on a *backward-looking, compounded-in-arrears* rate settles a payoff that depends on the realised path of overnight fixings across the whole accrual period, known only at the period's end. This is closer to an Asian option, its effective volatility is lower

than the corresponding forward-looking rate's spot volatility because the averaging inside the period smooths out noise, and in-arrears caplets trade at correspondingly different (generally lower, all else equal) implied volatilities than an otherwise-equivalent forward-looking caplet would.

Worked example 3.1 — why in-arrears convexity is not optional to model. Consider a 3-month caplet accrual period. Under a forward-looking convention, the rate is a single random draw, fixed on day one, exactly as in the classical LIBOR setting. Under a backward-looking, compounded-in-arrears convention, the payoff at the end of the period depends on the geometric compounding of roughly 63 daily overnight fixings observed *during* the period.

Step 1 — intuition via variance of an average. If daily overnight rate changes were independent with daily variance v, the variance of the simple arithmetic average of n such observations is v/n, dramatically smaller than the variance v of a single terminal draw. Compounding is not quite a simple average, but the qualitative effect survives: realized in-period variance is damped relative to a point-in-time observation.

Step 2 — consequence for the model. A model that treats an in-arrears SOFR caplet as if it were priced off a single terminal draw of the period-end rate — the naive LIBOR-era shortcut — systematically misprices the option, typically overstating its volatility and therefore its value, because it ignores the smoothing the daily compounding provides.

Interpretation. This is not a subtlety that only shows up in exotic products. It is present in every single SOFR cap and floor traded in the market today, which is the dominant form of vanilla rates optionality. Chapter 5 builds the in-arrears caplet payoff explicitly, and any BGM implementation aimed at today's market — rather than the LIBOR market every classic textbook was written for — must model the compounding convexity correctly or misprice its entire vanilla book.

ON THE DESK

The practical consequence for a calibration application: the "forward rate" object that BGM evolves cannot be taken naively off-the-shelf as "the SOFR rate." A production system needs to decide, and be explicit about, whether it is modelling a forward-looking term rate (behaves like old LIBOR, Black's formula applies cleanly) or a backward-looking compounded rate (needs an in-arrears adjustment, and its effective volatility must be calibrated to in-arrears market quotes, not forward-looking ones borrowed from a different convention). Mixing the two conventions inside one calibration — fitting to forward-looking vols but pricing in-arrears exotics with the fitted parameters, or vice versa — is a real and recurring implementation error.

3.4 What this means for the rest of the book

To keep the exposition focused on the model rather than on convention bookkeeping, most of this book's worked examples use a forward-looking term rate convention, which is mathematically identical to the classical LIBOR setting BGM was originally built for and lets Black's formula (Chapter 6) apply directly. The in-arrears adjustment from Worked Example 3.1 is revisited explicitly in Chapter 5, where the caplet payoff is built in full, and again in Chapter 20, where it is treated as a first-class requirement of a real calibration engine rather than a footnote.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Explain why discounting and forwarding curves became separate objects after the 2008 crisis, and what each one is used for.
2. Name the RFRs replacing LIBOR in USD, GBP and EUR, and state that they are overnight rates by construction.
3. Explain the structural difference between a forward-looking term rate and a backward-looking compounded-in-arrears rate, and why a caplet on the latter behaves like an Asian option rather than a plain vanilla option.